

and the function $\mathbb{1}_K c(a, b)$ is upper semi-continuous on $X \times Y$ (since it is continuous, positive and bounded on a closed set, and vanishes outside it). This implies

$$\limsup_n \int c(a, b) d\gamma_n \leq \int \mathbb{1}_K c(a, b) d\gamma + 2M\delta \leq \int c(a, b) d\gamma + 2M\delta$$

and, since δ is arbitrary, $\limsup_n \int c(a, b) d\gamma_n \leq \int c(a, b) d\gamma$. This proves upper semi-continuity of the integral functional when c is bounded by M . An analogous computation with $M - c$ instead of c proves lower semi-continuity.

If c is positive but unbounded, just approximate it from below with its truncations $c_M = c \wedge M$, and lower semi-continuity is proven for the integral functional, which will be a sup of lower semi-continuous functionals. By replacing c with the function $\tilde{X} \times \tilde{Y} \ni (\tilde{x}, \tilde{y}) \mapsto f(\tilde{x}) + g(\tilde{y}) - c(\tilde{x}, \tilde{y})$ upper semi-continuity is proven as well. $\square$ $\square$

Box 1.6. – Memo – Lusin's theorem.

A well-known theorem in measure theory states that every measurable function f on a reasonable measure space (X, μ) , is actually continuous on a set K with $\mu(X \setminus K)$ small. This set K can be taken compact. Actually, there can be at least two statements: either we want f to be merely continuous on K , or we want f to coincide on K with a continuous function defined on X . This theorem is usually stated for real-valued functions, but we happen to need it for functions valued in more general spaces. Let us be more precise: take a topological space X endowed with a finite regular measure μ (i.e. any Borel set $A \subset X$ satisfies $\mu(A) = \sup\{\mu(K) : K \subset A, K \text{ compact}\} = \inf\{\mu(B) : B \supset A, B \text{ open}\}$). The arrival space Y will be supposed to be second-countable (i.e. it admits a countable family $(B_i)_i$ of open sets such that any other open set $B \subset Y$ may be expressed as a union of B_i ; for instance, separable metric spaces are second-countable).

Theorem (weak Lusin) - Under the above assumptions on X, Y, μ , if $f : X \rightarrow Y$ is measurable, then for every $\varepsilon > 0$ there exists a compact set $K \subset X$ such that $\mu(X \setminus K) < \varepsilon$ and the restriction of f to K is continuous.

Proof - For every $i \in \mathbb{N}$, set $A_i^+ = f^{-1}(B_i)$ and $A_i^- = f^{-1}(B_i^c)$. Consider compact sets $K_i^\pm \subset A_i^\pm$ such that $\mu(A_i^\pm \setminus K_i^\pm) < \varepsilon 2^{-i}$. Set $K_i = K_i^+ \cup K_i^-$ and $K = \bigcap_i K_i$. For each i we have $\mu(X \setminus K_i) < \varepsilon 2^{1-i}$. By construction, K is compact and $\mu(X \setminus K) < 4\varepsilon$. To prove that f is continuous on K it is sufficient to check that $f^{-1}(B) \cap K$ is relatively open in K for each open set B , and it is enough to check this for $B = B_i$. Equivalently, it is enough to prove that $f^{-1}(B_i^c) \cap K$ is closed, and this is true since it coincides with $K_i^- \cap K$.

Theorem (strong Lusin) - Under the same assumptions on X , if $f : X \rightarrow \mathbb{R}$ is measurable, then for every $\varepsilon > 0$ there exists a compact set $K \subset X$ and a continuous function $g : X \rightarrow \mathbb{R}$ such that $\mu(X \setminus K) < \varepsilon$ and $f = g$ on K .

Proof - First apply weak Lusin's theorem, since $\mathbb{R}$ is second countable. Then we just need to extend $f|_K$ to a continuous function g on the whole X . This is possible since $f|_K$ is uniformly continuous (as a continuous function on a compact set) and hence has a modulus of continuity ω : $|f(x) - f(x')| \leq \omega(d(x, x'))$ (the function ω can be taken sub additive and continuous). Then define $g(x) = \inf\{f(x') + \omega(d(x, x')) : x' \in K\}$. It can be easily checked that g is continuous and coincides with f on K .

Note that this last proof strongly uses the fact that the arrival space is $\mathbb{R}$. It could be adapted to the case of $\mathbb{R}^d$ just by extending component-wise. On the other hand, it is

clear that the strong version of Lusin's Theorem cannot hold for any space Y : just take X connected and Y disconnected. A measurable function $f : X \rightarrow Y$ taking values in two different connected components on two sets of positive measure cannot be approximated by continuous functions in the sense of the strong Lusin's Theorem.

The consequence of all these continuity, semi-continuity, and compactness results is the existence, under very mild assumptions on the cost and the space, of an optimal transport plan γ . Then, if one is interested in the problem of Monge, the question may become “does this minimal γ come from a transport map T ?”. Should the answer to this question be yes, then (MP) would have a solution, which also solves a wider problem, that of minimizing among transport plans. This is the object of the next two sections 1.2 and 1.3. On the other hand, in some cases proving that the optimal transport plan comes from a transport map (or proving that there exists at least one optimal plan coming from a map) is equivalent to proving that (MP) has a solution, since very often the infimum among transport plans and among transport maps is the same. This depends on the presence of atoms (see Sections 1.4 and 1.5).

1.2 Duality

The problem (KP) is a linear optimization under convex constraints, given by linear equalities or inequalities. Hence, an important tool will be duality theory, which is typically used for convex problems. We will find a dual problem (DP) for (KP) and exploit the relations between dual and primal.

The first thing we will do is finding a formal dual problem, by means of an inf-sup exchange.

Let us express the constraint $\gamma \in \Pi(\mu, \nu)$ in the following way : note that, if $\gamma \in \mathcal{M}_+(X \times Y)$, then we have

$$\sup_{\varphi, \psi} \int_X \varphi d\mu + \int_Y \psi d\nu - \int_{X \times Y} (\varphi(x) + \psi(y)) d\gamma = \begin{cases} 0 & \text{if } \gamma \in \Pi(\mu, \nu), \\ +\infty & \text{otherwise,} \end{cases}$$

where the supremum is taken among bounded and continuous functions φ, ψ .

Hence, we can remove the constraints on γ if we add the previous sup, since if they are satisfied nothing has been added and if they are not we get $+\infty$ (which will be avoided by the minimization). Hence we may look at the problem we get:

$$\min_{\gamma} \int_{X \times Y} c d\gamma + \sup_{\varphi, \psi} \int_X \varphi d\mu + \int_Y \psi d\nu - \int_{X \times Y} (\varphi(x) + \psi(y)) d\gamma \quad (1.3)$$

and consider interchanging sup and inf:

$$\sup_{\varphi, \psi} \int_X \varphi d\mu + \int_Y \psi d\nu + \inf_{\gamma} \int_{X \times Y} (c(x, y) - (\varphi(x) + \psi(y))) d\gamma.$$

We would like the two above optimization problems (“inf sup” and “sup inf”) to be equivalent, and the value to be the same. This is not always possible, and the main tool to do it is a theorem by Rockafellar (see [260], Section 37) requiring concavity in one variable, convexity in the other one, and some compactness assumption. Yet, Rockafellar’s statement concerns finite-dimensional spaces, which is not the case here. To handle infinite-dimensional situations one needs to use a more general mini-max theorems².

For now, we prefer not to investigate anymore the question of obtaining the duality equality. The result is true (under suitable assumptions) and we will see later on how to prove it. For the moment, let us accept it as true.

If we come back to the maximization over (φ, ψ) , one can re-write the inf in γ as a constraint on φ and ψ :

$$\inf_{\gamma \geq 0} \int_{X \times Y} (c - \varphi \oplus \psi) d\gamma = \begin{cases} 0 & \text{if } \varphi \oplus \psi \leq c \text{ on } X \times Y \\ -\infty & \text{otherwise} \end{cases},$$

where $\varphi \oplus \psi$ denotes the function defined through $(\varphi \oplus \psi)(x, y) := \varphi(x) + \psi(y)$. The above equality is easy to see: if $\varphi \oplus \psi > c$ somewhere, then use measures γ concentrated on the set where this strict inequality holds, with mass tending to ∞ , and the integral tends to $-\infty$. This leads to the following dual optimization problem.

Problem 1.9. Given $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$ and the cost function $c : X \times Y \rightarrow [0, +\infty[$ we consider the problem

$$(DP) \quad \max \left\{ \int_X \varphi d\mu + \int_Y \psi d\nu : \varphi \in C_b(X), \psi \in C_b(Y) : \varphi \oplus \psi \leq c \right\}. \quad (1.4)$$

First of all, we notice that $\sup(DP) \leq \min(KP)$: it is enough to integrate the condition $\varphi \oplus \psi \leq c$ according to γ , to get

$$\int_X \varphi d\mu + \int_Y \psi d\nu = \int_{X \times Y} (\varphi \oplus \psi) d\gamma \leq \int_{X \times Y} c d\gamma.$$

This is valid for every admissible (φ, ψ) and every admissible γ , and proves the desired inequality.

Yet, (DP) does not admit a straightforward existence result, since the class of admissible functions lacks compactness. Let us recall the main result concerning compactness in the space of continuous functions.

Box 1.7. – Memo – Compactness for the uniform convergence.

Theorem (Ascoli-Arzelà) - If X is a compact metric space and $f_n : X \rightarrow \mathbb{R}$ are equi-continuous (i.e. for every $\varepsilon > 0$ there exists a common $\delta > 0$ such that $|f_n(x) - f_n(y)| < \varepsilon$ for all pairs x, y with $d(x, y) < \delta$ and for all n) and equi-bounded (i.e. there is a common

²We will give a proof in this spirit in Section 1.6.3.

constant C with $|f_n(x)| \leq C$ for all $x \in X$ and all n , then the sequence (f_n) admits a subsequence f_{n_k} uniformly converging to a continuous function $f : X \rightarrow \mathbb{R}$.

Conversely, a subset of $C(X)$ is relatively compact for the uniform convergence (if and only if its elements are equi-continuous and equi-bounded).

The same result is true if the arrival space $\mathbb{R}$ and the equiboundedness assumption are replaced with an arrival space which is a compact metric space.

Definition 1.10. Given a function $\chi : X \rightarrow \overline{\mathbb{R}}$ we define its c -transform (also called c -conjugate function) $\chi^c : Y \rightarrow \overline{\mathbb{R}}$ by

$$\chi^c(y) = \inf_{x \in X} c(x, y) - \chi(x).$$

We also define the $\bar{c}$ -transform of $\zeta : Y \rightarrow \overline{\mathbb{R}}$ by

$$\zeta^{\bar{c}}(x) = \inf_{y \in Y} c(x, y) - \zeta(y).$$

Moreover, we say that a function ψ defined on Y is $\bar{c}$ -concave if there exists χ such that $\psi = \chi^c$ (and, analogously, a function ϕ on X is said to be c -concave if there is $\zeta : Y \rightarrow \overline{\mathbb{R}}$ such that $\phi = \zeta^{\bar{c}}$) and we denote by $c\text{-conc}(X)$ and $\bar{c}\text{-conc}(Y)$ the sets of c - and $\bar{c}$ -concave functions, respectively (when $X = Y$ and c is symmetric this distinction between c and $\bar{c}$ will play no more any role and will be dropped as soon as possible).

It is important to note that the notion of c -concavity implies a bound on the modulus of continuity.

Box 1.8. – Memo – Continuity of functions defined as an inf or sup

Proposition - Let $(f_\alpha)_\alpha$ be a family (finite, infinite, countable, uncountable...) of functions all satisfying the same condition

$$|f_\alpha(x) - f_\alpha(x')| \leq \omega(d(x, x')).$$

Consider f defined through $f(x) := \inf_\alpha f_\alpha(x)$. Then f also satisfies the same estimate.

This can be easily seen from $f_\alpha(x) \leq f_\alpha(x') + \omega(d(x, x'))$, which implies $f(x) \leq f_\alpha(x') + \omega(d(x, x'))$ since $f \leq f_\alpha$. Then, taking the infimum over α at the r.h.s. one gets $f(x) \leq f(x') + \omega(d(x, x'))$. Interchanging x and x' one obtains

$$|f(x) - f(x')| \leq \omega(d(x, x')).$$

In particular, if the function $\omega : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfies $\lim_{t \rightarrow 0} \omega(t) = 0$ (which means that the family $(f_\alpha)_\alpha$ is equicontinuous), then f has the same modulus of continuity (i.e. the same function ω) as the functions f_α . The same idea obviously work for the supremum instead of the infimum.

In our case, if c is continuous and finite on a compact set, and hence uniformly continuous, this means that there exists an increasing continuous function $\omega : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $\omega(0) = 0$ such that

$$|c(x, y) - c(x', y')| \leq \omega(d(x, x') + d(y, y')).$$

Hence, when we take the definition of χ^c , we have $\chi^c(y) = \inf_x g_x(y)$ with $g_x(y) := c(x, y) - \chi(x)$ and the functions g_x satisfy $|g_x(y) - g_x(y')| \leq \omega(d(y, y'))$. This proves that χ^c shares the same continuity modulus also.

It is quite easy to realize that, given a pair (φ, ψ) in the maximization problem (DP), one can always replace it with (φ, φ^c) , and then with $(\varphi^{c\bar{c}}, \varphi^c)$, and the constraints are preserved and the integrals increased. Actually one could go on, but we will see that $\varphi^{c\bar{c}c} = \varphi^c$ for any φ (see Section 1.6). The goal of these transformations is to “improve” the maximizing sequence so as to get a uniform bound on its continuity.

A consequence of these considerations is the following existence result.

Proposition 1.11. *Suppose that X and Y are compact and c is continuous. Then there exists a solution (φ, ψ) to problem (DP) and it has the form $\varphi \in c\text{-conc}(X)$, $\psi \in \bar{c}\text{-conc}(Y)$, and $\psi = \varphi^c$. In particular*

$$\max(\text{DP}) = \max_{\varphi \in c\text{-conc}(X)} \int_X \varphi \, d\mu + \int_Y \varphi^c \, d\nu.$$

Proof. From the considerations above we can take a maximizing sequence (φ_n, ψ_n) and improve it, by means of c - and $\bar{c}$ -transforms, so that we can assume a uniform bound on the continuity of these functions (the same modulus of continuity as c). Instead of renaming the sequence, we will still call (φ_n, ψ_n) the new sequence obtained after these transforms. We only need to check equi-boundedness so as to apply Ascoli-Arzelà’s theorem. This may be done if we note that adding a constant to φ and subtracting it to ψ is always possible: the value of the functional does not change, nor the constraints are affected. Hence, since φ_n is continuous on a compact set and hence bounded, we can always subtract its minimum and suppose without loss of generality that $\min \varphi_n = 0$. We get $\max \varphi_n \leq \omega(\text{diam} X)$ (since the oscillation of a function is always less than its modulus of continuity computed at the highest possible distance in the set). So, if we have chosen $\psi_n = \varphi_n^c$, we also have $\psi_n(y) = \inf_x c(x, y) - \varphi_n(x) \in [\min c - \omega(\text{diam} X), \max c]$. This gives uniform bounds on φ_n and ψ_n and allows to apply Ascoli-Arzelà’s theorem.

Passing to a subsequence we can assume $\varphi_n \rightarrow \varphi$ and $\psi_n \rightarrow \psi$, both convergences being uniform. It is easy to see that

$$\int_X \varphi_n \, d\mu + \int_Y \psi_n \, d\nu \rightarrow \int_X \varphi \, d\mu + \int_Y \psi \, d\nu,$$

as a consequence of uniform convergence. Moreover

$$\varphi_n(x) + \psi_n(y) \leq c(x, y) \Rightarrow \varphi(x) + \psi(y) \leq c(x, y)$$

(here pointwise convergence would have been enough). This shows that (φ, ψ) is an admissible pair for (DP) and that it is optimal. $\square$ $\square$

If we admit the duality result $\min(\text{KP}) = \max(\text{DP})$ (the proof is postponed to Section 1.6), then we also have

$$\min(\text{KP}) = \max_{\varphi \in c\text{-conc}(X)} \int_X \varphi \, d\mu + \int_Y \varphi^c \, d\nu,$$

which also shows that the minimum value of (KP) is a convex function of (μ, ν) , as it is a supremum of linear functionals.

Definition 1.12. The functions φ realizing the maximum in (3.1) are called *Kantorovich potentials* for the transport from μ to ν . This is in fact a small abuse, because traditionally this term was used only in the case $c(x, y) = |x - y|$, but it is nowadays understood in the general case as well.

Remark 1.13. Note that Kantorovich potentials are not necessarily c -concave functions, but coincide with the c -concave function $\varphi^{c^c} \mu$ -a.e. However, it is always possible to choose a c -concave Kantorovich potential, and we will do it often.

1.3 The case $c(x, y) = h(x - y)$ for h strictly convex, and the existence of an optimal T

This section is mainly devoted to the results that we can obtain in the case where $X = Y = \Omega \subset \mathbb{R}^d$ and the cost c is of the form $c(x, y) = h(x - y)$, for a strictly convex function h . We will also assume Ω to be compact for simplicity. This case allows for very strong results, and in particular we will find existence, as well as a representation formula, for the optimal T. Anyway, the first few lines of this section will be concerned with a more general case: that of costs functions c satisfying a *twist condition*, the most remarkable case being exactly those of the form $h(x - y)$ with h strictly convex.

The main tool is the duality result. More precisely, we use Theorem 1.39, which applies to the case of compact domains with continuous cost functions c . It guarantees $\max(\text{DP}) = \min(\text{KP})$.

From the equality between the minimum of (KP) and the maximum of (DP) and the fact that both extremal values are realized, one can consider an optimal transport plan γ and a Kantorovich potential φ and write

$$\varphi(x) + \varphi^c(y) \leq c(x, y) \text{ on } \Omega \times \Omega \text{ and } \varphi(x) + \varphi^c(y) = c(x, y) \text{ on } \text{spt}(\gamma).$$

The equality on $\text{spt}(\gamma)$ is a consequence of the inequality which is valid everywhere and of

$$\int_{\Omega \times \Omega} c \, d\gamma = \int_{\Omega} \varphi \, d\mu + \int_{\Omega} \varphi^c \, d\nu = \int_{\Omega \times \Omega} (\varphi(x) + \varphi^c(y)) \, d\gamma,$$

which implies equality γ -a.e. These functions being continuous, the equality is satisfied on a closed set, i.e. on the support of the measure γ (let us recall the definition of support of a measure, not to be confused with sets where it is concentrated).